

ALGEBRA QUALIFYING EXAMINATION

Winter 2024

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- Part I is asking for statements of some definitions/theorems. Three questions, 6 points each, give concise but accurate answers.
- Part II is True/False. Five questions, 8 points each, give a sketch proof or counterexample for full credit.
- Part III is longer problems. Three questions, 14 points each, give concise but clear proofs stating any significant results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following (6 points each):

1. The definition of the determinant of an endomorphism $f : V \rightarrow V$ of a finite-dimensional vector space V .
2. Two different but equivalent definitions of an injective module.
3. The Krull-Schmidt Theorem for modules of finite length.

Part II. True or false? If true provide a brief explanation, if false provide a counterexample (8 points each):

1. The dihedral group D_{2024} of order 4048 is nilpotent.
2. Any semisimple $\mathbb{R}$ -algebra of dimension 3 is commutative.
3. If $\mathcal{A}$ is a groupoid (= a category in which every morphism is an isomorphism) and $\mathcal{B}$ is a category that is equivalent to $\mathcal{A}$ then $\mathcal{B}$ is a groupoid.
4. The algebraic set $\mathcal{V}(x^2 + y^2 + z^2 - 1) \subset \mathbb{A}_{\mathbb{C}}^3$ is reducible.
5. Let χ be the character of the symmetric group S_4 that is induced from the trivial character of the subgroup S_3 . The inner product (χ, χ) equals 2.

Part III. Give complete answers to the following (14 points each):

1. Let F be a field of characteristic $p > 0$ and consider the polynomial

$$f(x) := x^p - x - c \in F[x]$$

for some $c \in F$. Let α be a root of $f(x)$ in an algebraic closure $\overline{F}$ of F .

- (a) Prove that $f(x)$ has roots $\alpha, \alpha + 1, \dots, \alpha + p - 1$.
- (b) Prove that $F(\alpha)/F$ is a Galois extension.
- (c) Prove that $f(x)$ is either irreducible or splits into linear factors in $F[x]$.
(*Hint:* Assume that $\alpha \notin F$ and construct some elements of $\text{Gal}(F(\alpha)/F)$.)

2. Let R and S be rings, U be a left R -module, V be an (R, S) -bimodule and W be a left S -module.

- (a) Construct a homomorphism of Abelian groups

$$\alpha_U : \text{Hom}_R(U, V) \otimes_S W \rightarrow \text{Hom}_R(U, V \otimes_S W)$$

which is natural in U , such that α_R is an isomorphism (here, R is the left regular R -module). You should explain carefully why your homomorphism is well defined, but there is no need to justify the naturality.

- (b) Prove that α_U is an isomorphism when U is a finitely generated projective left R -module.

3. A *non-trivial idempotent* in a commutative ring R is an element $e \in R - \{0, 1\}$ such that $e^2 = e$. Note that $e(1 - e) = 0$, hence, non-trivial idempotents are (non-zero) zero divisors.

- (a) Prove that the ring $R := \mathbb{C}[x, y]/(xy)$ contains a (non-zero) zero divisor but has no non-trivial idempotents¹.
- (b) Continue with $R = \mathbb{C}[x, y]/(xy)$ as in (a), and let S be the multiplicative set $1, f, f^2, \dots$ where $f = x + y$. Find a non-trivial idempotent in the localization $S^{-1}R = R_f$.

¹Geometrically, this means that the algebraic set $\mathcal{V}(xy)$ is *connected*.